External side effects and recovery

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete external-effect recovery decision record.

Keep stable identity, status evidence, uncertainty, read permission, resend conditions and separate compensation authority together.

Keep one habit: Recall → W1–W3/W5 → independent uncertainty case → project recovery record → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

- The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

  Preserve **compensation** as a separate authorised action.

Timeouts retries and cancellation

> **My remembered idea, example and correction**

Planning and action selection

> **My remembered idea, example and correction**

Failure budgets and service objectives

> **My remembered idea, example and correction**

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fixture BOOK-1 is reliably committed with existing booking result B1, but its acknowledgement was lost. Reconcile by original operation ID and return the existing result to an authorized caller; do not create a second effect. A prepared state is not confirmed failure. Unknown/unavailable or unreliable/mismatched status remains review/uncertainty.

  Separate printer J6 has an unknown outcome after a lost reply; reliable completion returns its existing result, while reliable absence changes evidence but does not itself grant resend permission. Only separately trusted same-key resend authority with required provider duplicate controls yields a candidate; no resend executes here.

  Compensation such as cancellation is another action with its own exact approval and new operation identity, preserving original history. The local helper assumes supplied authoritative/access flags and classroom approval equality, copies returned results and leaves original records unchanged. It performs no real status lookup, print, booking, transfer or durable/exactly-once recovery.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fixture BOOK-1 is reliably committed with existing booking result B1, but its acknowledgement was lost. Reconcile by original operation ID and return the existing result to an authorized caller; do not create a second effect. A prepared state is not confirmed failure. Unknown/unavailable or unreliable/mismatched status remains review/uncertainty.

  Separate printer J6 has an unknown outcome after a lost reply; reliable completion returns its existing result, while reliable absence changes evidence but does not itself grant resend permission. Only separately trusted same-key resend authority with required provider duplicate controls yields a candidate; no resend executes here.

  Compensation such as cancellation is another action with its own exact approval and new operation identity, preserving original history. The local helper assumes supplied authoritative/access flags and classroom approval equality, copies returned results and leaves original records unchanged. It performs no real status lookup, print, booking, transfer or durable/exactly-once recovery.

- A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fixture BOOK-1 is reliably committed with existing booking result B1, but its acknowledgement was lost. Reconcile by original operation ID and return the existing result to an authorized caller; do not create a second effect. A prepared state is not confirmed failure. Unknown/unavailable or unreliable/mismatched status remains review/uncertainty.

  Separate printer J6 has an unknown outcome after a lost reply; reliable completion returns its existing result, while reliable absence changes evidence but does not itself grant resend permission. Only separately trusted same-key resend authority with required provider duplicate controls yields a candidate; no resend executes here.

  Compensation such as cancellation is another action with its own exact approval and new operation identity, preserving original history. The local helper assumes supplied authoritative/access flags and classroom approval equality, copies returned results and leaves original records unchanged. It performs no real status lookup, print, booking, transfer or durable/exactly-once recovery.

- A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main fixture BOOK-1 is reliably committed with existing booking result B1, but its acknowledgement was lost. Reconcile by original operation ID and return the existing result to an authorized caller; do not create a second effect. A prepared state is not confirmed failure. Unknown/unavailable or unreliable/mismatched status remains review/uncertainty.

  Separate printer J6 has an unknown outcome after a lost reply; reliable completion returns its existing result, while reliable absence changes evidence but does not itself grant resend permission. Only separately trusted same-key resend authority with required provider duplicate controls yields a candidate; no resend executes here.

  Compensation such as cancellation is another action with its own exact approval and new operation identity, preserving original history. The local helper assumes supplied authoritative/access flags and classroom approval equality, copies returned results and leaves original records unchanged. It performs no real status lookup, print, booking, transfer or durable/exactly-once recovery.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled external side effects and recovery **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of external side effects and recovery.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for External side effects and recovery. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

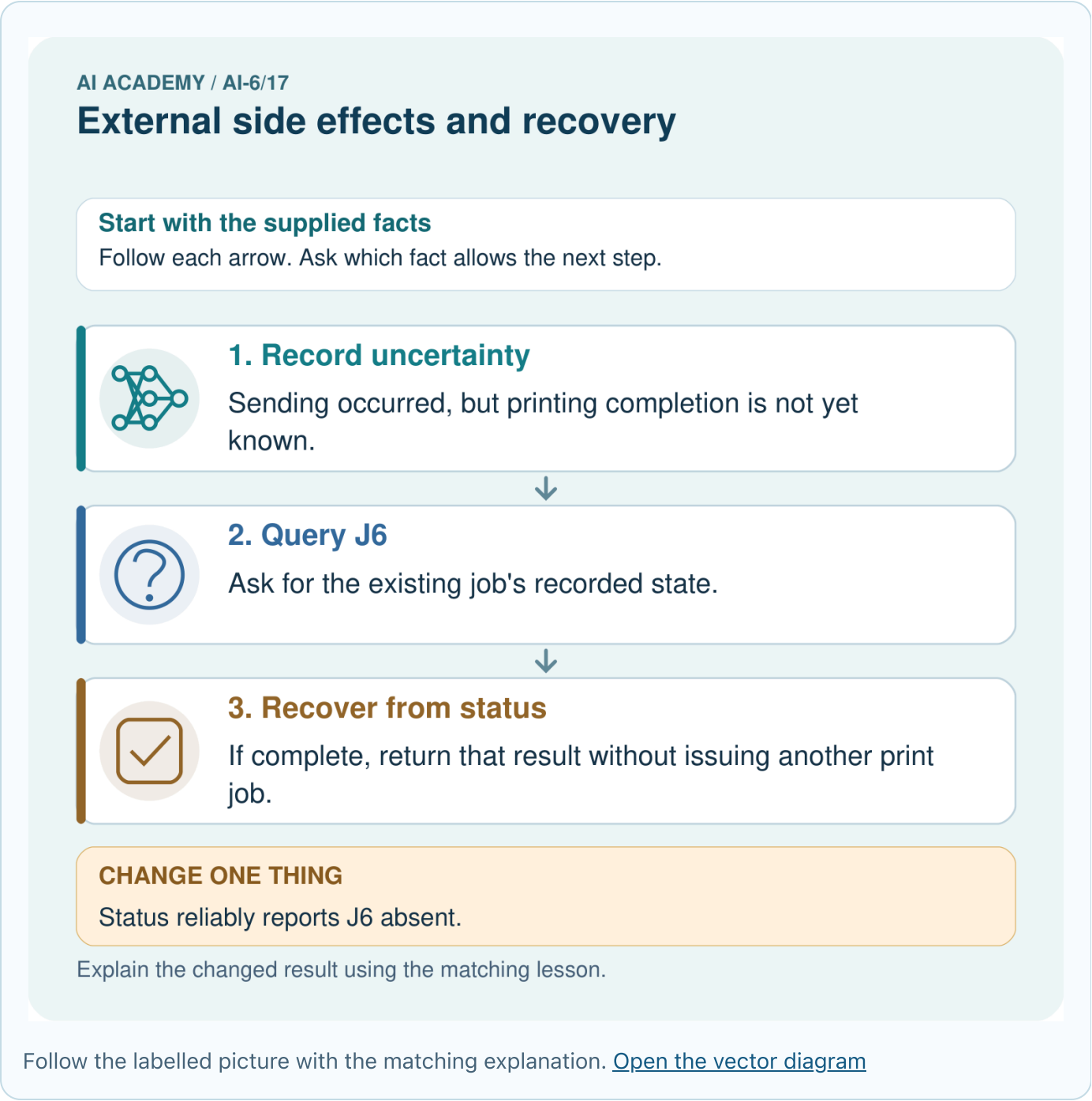
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Recover an uncertain print job. A badge printer loses its reply after receiving a print request.
- Job key J6 identifies one badge. The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.
- Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Recover an uncertain print job. A badge printer loses its reply after receiving a print request.
- Which labelled step matches this detail? If complete, return that result without issuing another print job.
- 1. Record uncertainty
- 2. Recover from status
- 3. Query J6

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Recover from status If complete, return that result without issuing another print job. A lost response does not show whether an external effect occurred.

Recovery distinguishes unknown outcome from a confirmed failed action.

Changed-case question and optional hint

- Changed-case practice: Status reliably reports J6 absent. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. A policy-approved resend using the same deduplication key may be considered.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Repair and retain the record

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

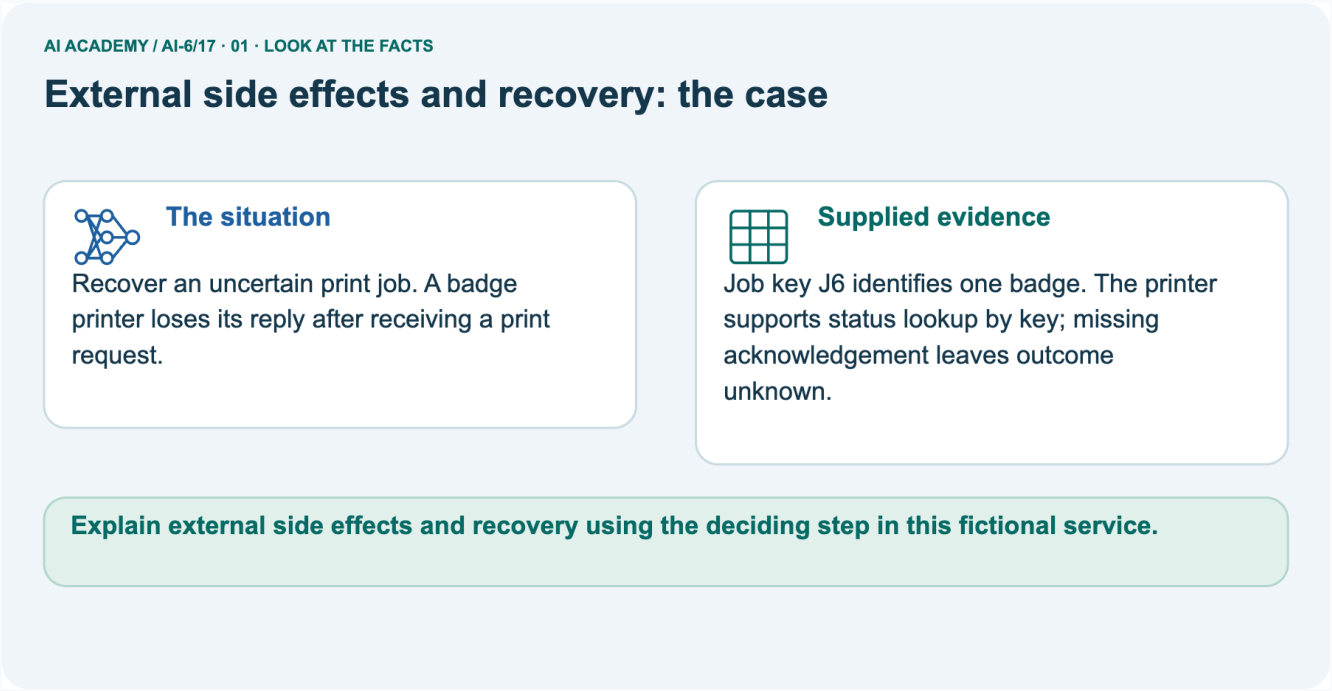
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. External side effects and recovery: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

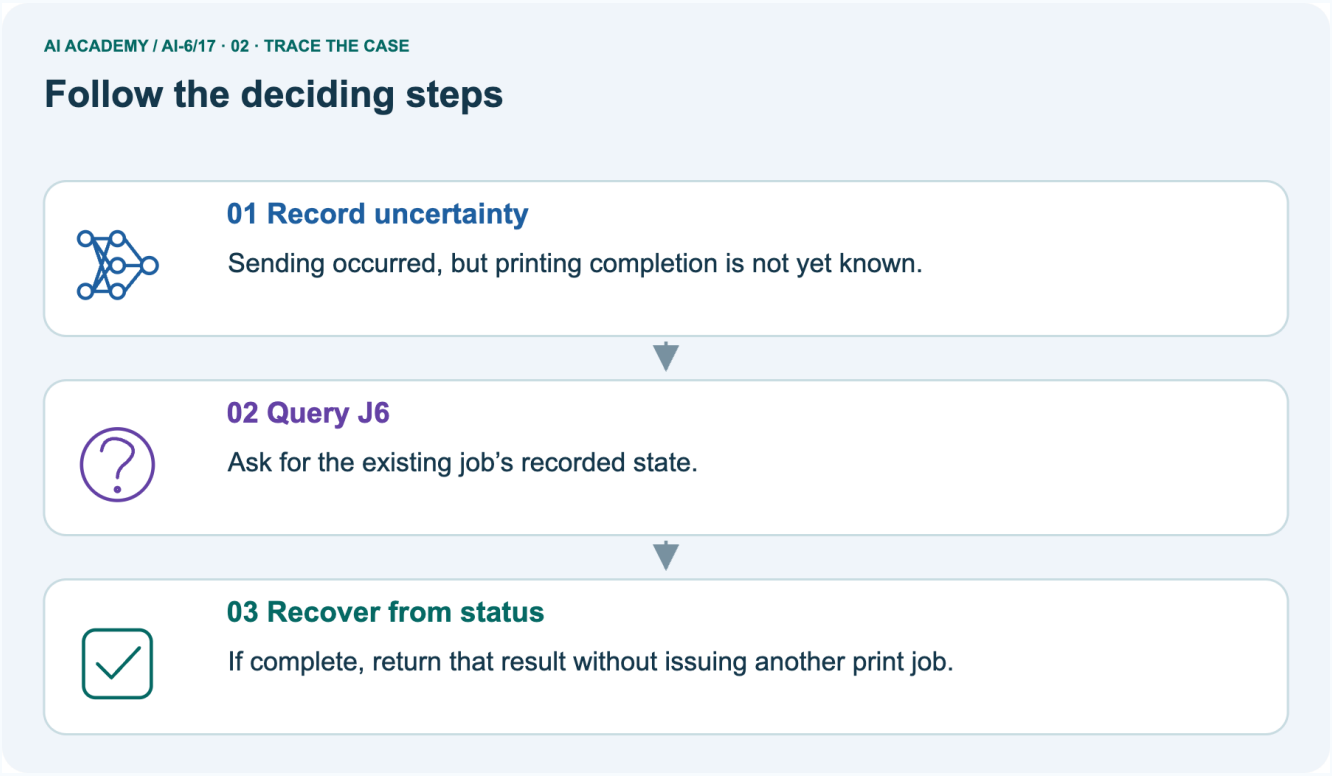
Recover an uncertain print job. A badge printer loses its reply after receiving a print request. Supplied facts: Job key J6 identifies one badge.

The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



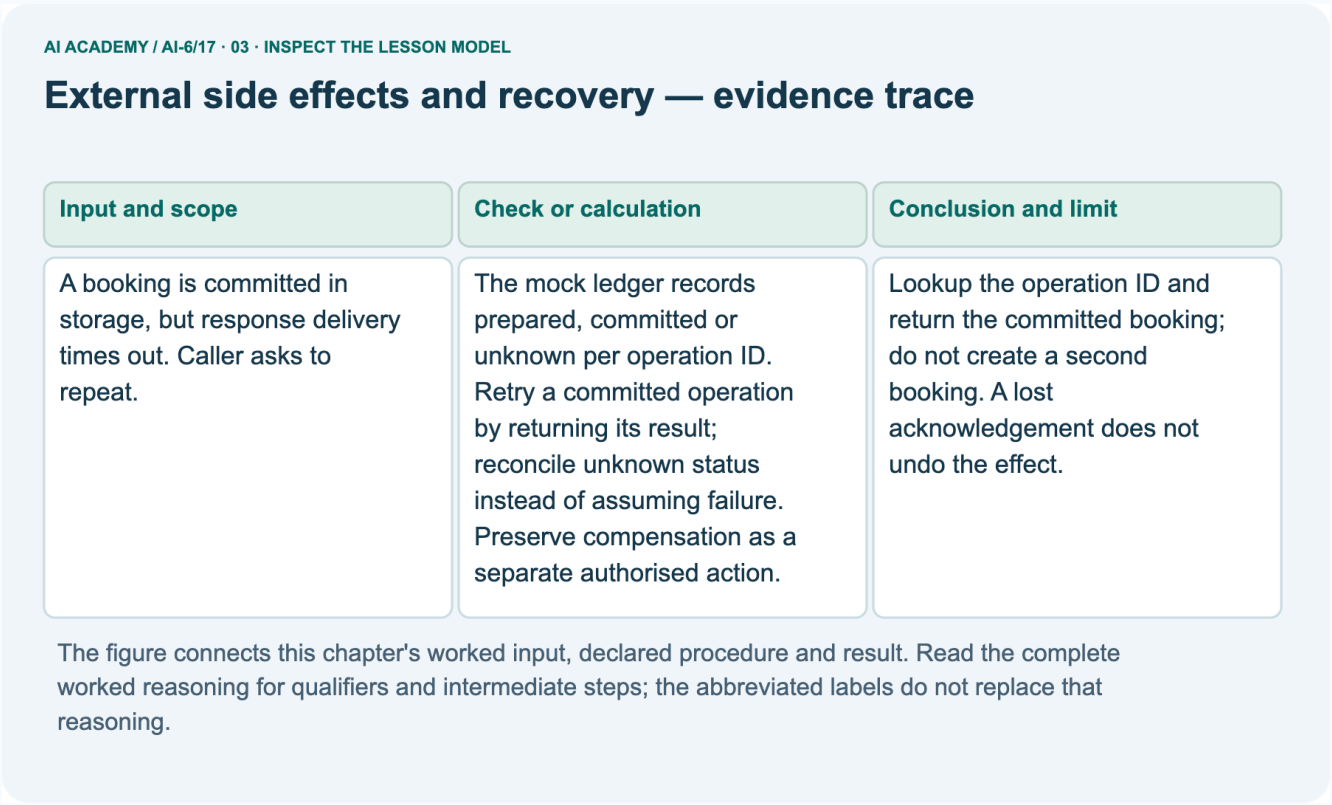
Follow the labelled picture with the matching explanation. Open the vector diagram

Record uncertainty: Sending occurred, but printing completion is not yet known. → Query J6: Ask for the existing job's recorded state. → Recover from status: If complete, return that result without issuing another print job.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. External side effects and recovery — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

External side effects and recovery — **evidence** trace. A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID.

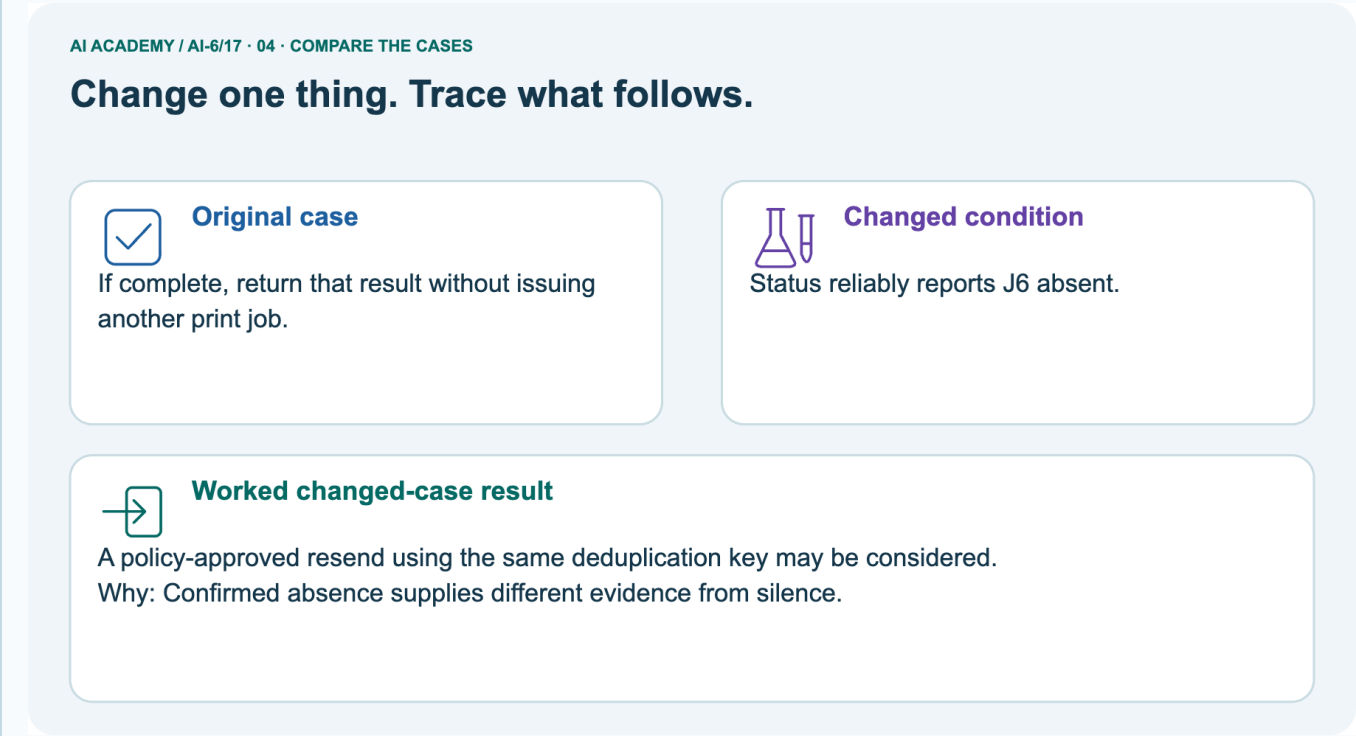
Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

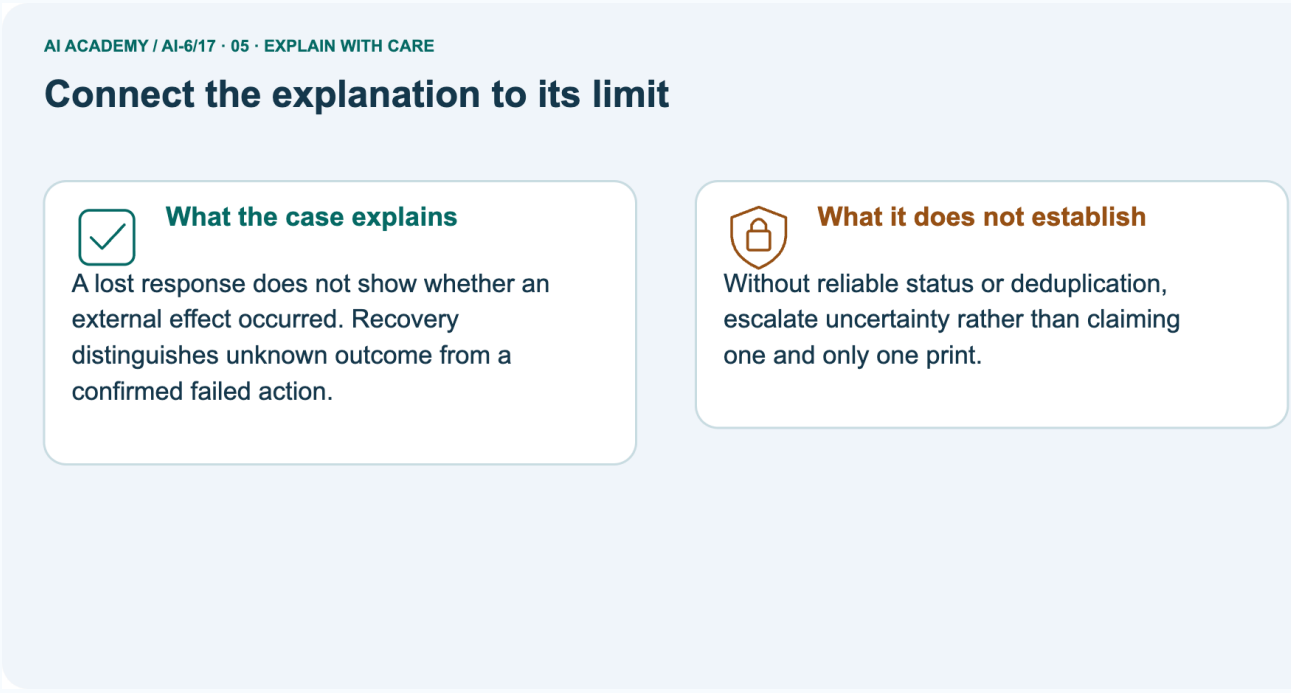
Original case: If complete, return that result without issuing another print job. Changed condition: Status reliably reports J6 absent.

Expected result: A policy-approved resend using the same deduplication key may be considered. Reason: Confirmed absence supplies different **evidence** from silence.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

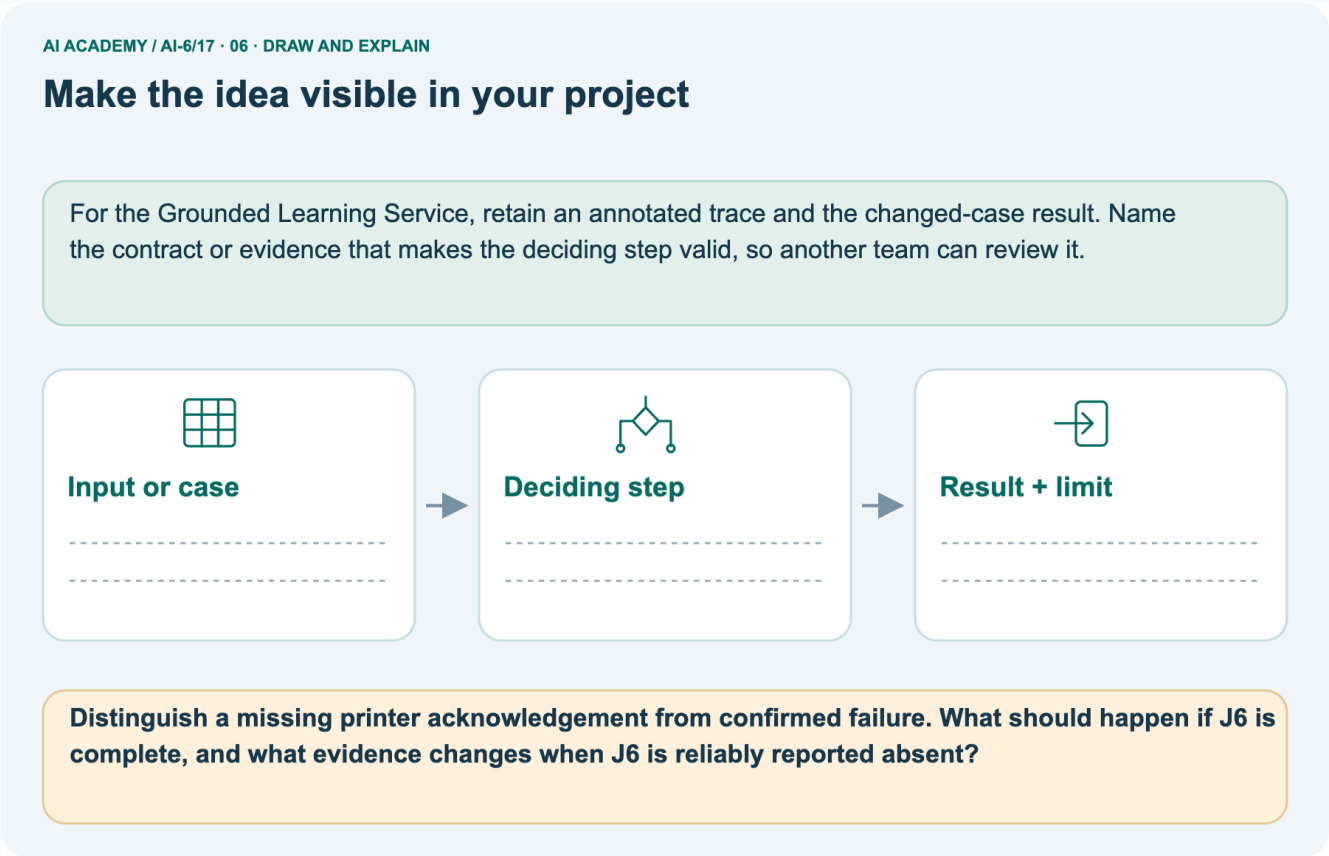
A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Limit: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Distinguish a missing printer **acknowledgement** from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ **Earlier wording, alternatives and picture transcripts**

01. External side effects and recovery: the case

02. Follow the deciding steps

03. External side effects and recovery — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.